

A4 · Where Does This Datagram Go?

CS425 Computer Networks · instructor led · every command runs on Onyx (`ssh onyx`)

YOUR NOTES, BY HAND
turn in at the end of class Thursday
graded pass/fail

Name _____ Date _____

Follow along on Onyx and take notes as we go. For each round write what you **expected** before we ran it, what you **saw**, and **why** it came out that way. Sketches, binary, and copied output all count. These are your notes for the exam, so write what you will want to read later.

Round 1 · Onyx's address

`ip -4 -br addr`

Onyx's prefix in binary, the network and broadcast addresses, the usable hosts, and whether it is private.

Round 2 · longest prefix match

`ip route · ip route get`

Which line each destination matched, direct or through the gateway, and where `127.0.0.1` came from.

Round 3 · how big can a datagram be?

`ping -c 1 -M do -s SIZE`

The MTU, the biggest payload that fit, where the other 28 bytes went, and who fragments without `-M do`.

Round 4 · counting routers with TTL

`tracert -n mirrors.xmission.com`

How many hops, how TTL revealed each router, and what the `no reply` hops mean.

Exit · beyond the destination address

One other header field a middlebox could match on, and what it might do with a match.
